

FTC Autonomous Answer Key

Part 9 — Drive Forward Three Feet and Stop

Use with the Part 9 student workbook and Java example

Accept equivalent wording when calculations and safety meaning remain correct.

Calculation answers

ITEM	WORK	ANSWER
Wheel diameter in inches	$96 / 25.4$	3.7795 in
Circumference	$3.7795 \times \pi$	11.874 in
Counts per inch	$537.7 / 11.874$	45.2848
36-inch target	36×45.2848	1,630.25
Whole-count target	round	1,630 counts

Code order

- 1. Map left and right motors.
- 2. Set directions and BRAKE.
- 3. Reset encoders.
- 4. waitForStart.
- 5. Set both target positions.
- 6. Select RUN_TO_POSITION.
- 7. Apply power 0.35.
- 8. Monitor isBusy and timeout.
- 9. Command zero motor power.

Stop paths

Normal completion, Driver Station STOP, or timeout all lead to zero motor power. The finally block reinforces the zero-power cleanup.

Calibration reasoning

new CPI = old CPI x commanded distance / actual distance. If the robot travels short, the denominator is below 36, so the new CPI increases. If it travels long, CPI decreases.

Example: $45.2848 \times 36 / 34.5 = 47.2537$ counts per inch, rounded appropriately for the team's constant and evidence.

Expected explanations

TOPIC	EXPECTED IDEA
Encoder	Measures motor rotation, not direct floor position.
RUN_TO_POSITION	Uses encoder feedback to move toward a target count.
Timeout	Backup exit if the target is not reached; not the distance measurement.
BRAKE	Reduces coasting at zero power; does not lock the robot.
Repeated trials	Separate consistent scale error from random slip or mechanical inconsistency.
Safe approach	Requires STOP, visible DISABLED, and safety-captain permission.

Strong exit response

A complete answer uses wheel circumference to convert 537.7 counts per revolution into about 45.28 counts per inch, multiplies by 36 for about 1,630 target counts, and explains that slip can change real travel. It names at least one software stop path and the human sequence: clear the lane, POWERING ON, pause, confirm clear, press STOP, and verify DISABLED before approach.